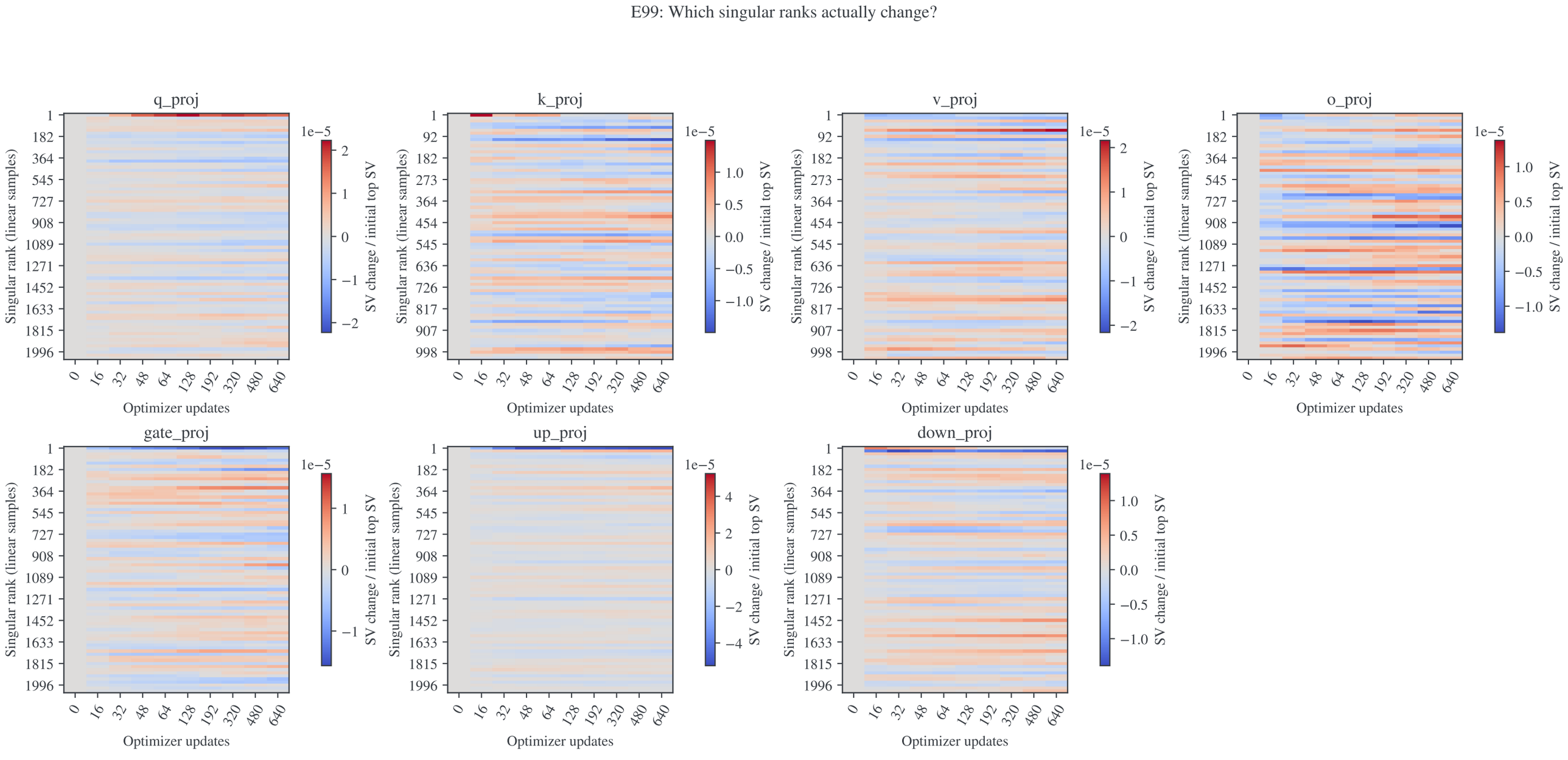

E99: Which singular ranks actually change?
q_proj
k_proj
v_proj
o_proj
gate_proj
up_proj
down_proj
Singular rank (linear samples)
Optimizer updates
SV change / initial top SV
1e−5
196/196 matrices; 28 full layers. Representative layer 14; 80 exact rank samples. Signed color scales are panel-specific.